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Abstract

This paper looks at the cybersecurity risks facing vehicle navigation systems, which have become a core part of modern transportation infrastructure. The focus is on four main areas: GPS spoofing, sensor-level attacks, encryption weaknesses, and insecure communication protocols. For each of these areas I try to explain not just what the threat is, but how it works in practice, what real-world incidents have occurred, and what the most effective defenses are. The paper also touches on future developments post-quantum cryptography, AI-based detection systems, and blockchain that are likely to change the security picture significantly over the coming years. My overall conclusion is that the problem cannot be solved by any single technology; it requires a layered approach that treats security as a design requirement from the start, not something patched in later.

1. Introduction

When I first started reading about vehicle navigation security, I was surprised by how wide the gap is between how reliable people assume GPS to be and how easy it is to manipulate. Most drivers, and even many engineers outside the field, tend to treat GPS as a solid, trustworthy reference. In reality, the civilian GPS signal is unencrypted, arrives at the receiver with a power level of roughly −130 dBm (about twenty billion times weaker than a typical Wi-Fi signal), and contains no mechanism for the receiver to verify that the signal genuinely came from a satellite rather than a transmitter sitting on a rooftop nearby.

Navigation systems have evolved enormously since the first commercial GPS receivers appeared in the late 1980s. The Global Navigation Satellite System (GNSS) family now includes the American GPS, the Russian GLONASS, the European Galileo, and the Chinese BeiDou, all of which provide Positioning, Navigation, and Timing (PNT) services to billions of users. Modern vehicles layer additional sensors on top of GNSS-accelerometers, gyroscopes, cameras, LiDAR, and radar-and connect to the outside world through Bluetooth, Wi-Fi, cellular, and dedicated Vehicle-to-Everything (V2X) links. The result is a system that is far more accurate and capable than anything that existed a generation ago, but also one that exposes a much larger attack surface.

Aviation relies on GNSS for instrument approaches in low visibility; container ships use it to navigate busy straits without grounding; ambulances depend on it to reach emergencies faster;

and the emerging class of autonomous vehicles needs it, along with onboard sensor data, to make safety-critical decisions at highway speed. The stakes of getting the security wrong are therefore very high not just financially, but in terms of human safety.

2. Overview of Vehicle Navigation Systems

It is worth spending a moment on the basic architecture before diving into the threats, because the vulnerabilities in navigation systems are not random, they map directly onto specific components and the interfaces between them.

The core of any GNSS-based navigation system is the receiver, which listens for timing pulses broadcast by satellites orbiting at around 20,000 km altitude. Each satellite transmits a pseudorandom noise (PRN) code that the receiver uses to calculate the signal travel time; with four or more satellites visible, the receiver can solve for its three-dimensional position and its own clock error simultaneously. Under open-sky conditions a modern receiver can achieve horizontal accuracy of about 1–3 metres, and with differential corrections from a ground station or a satellite-based augmentation system (SBAS), that can improve to the decimetre level.

The problem is that open-sky conditions are the exception in urban driving. Buildings block and reflect satellite signals, a phenomenon called multipath, which degrades accuracy and can even cause the receiver to lose lock entirely. This is where the Inertial Navigation System (INS) becomes important. An INS contains accelerometers that measure the vehicle's acceleration along each axis and gyroscopes that measure rotation rates; by integrating these measurements forward from a known starting point, it can propagate a position estimate without any external reference. The drawback is that integration errors accumulate over time, so a pure INS solution drifts and needs to be corrected periodically by GNSS or some other absolute reference.

Beyond GNSS and INS, modern vehicles add cameras for lane detection and traffic sign recognition, LiDAR for three-dimensional obstacle mapping, radar for measuring relative velocity to nearby objects, and magnetometers for heading reference. All of these sensors feed into a central processing unit-or in more complex vehicles, a network of Electronic Control Units (ECUs)-that fuses the data and produces the navigation solution that the driver or the automated system acts on. Communication links to the outside world (cellular networks, DSRC roadside units, cloud map servers)

add further interfaces, each of which represents a potential point of entry for an attacker. The overall picture is one of considerable complexity, and the security of the whole system is only as strong as its weakest component.

3. GPS Spoofing

3.1 How It Works

GPS spoofing is, at its core, a signal-level deception attack. The attacker generates counterfeit GNSS signals that replicate the structure of genuine satellite transmissions-the same carrier frequency (L1 at 1575.42 MHz for GPS), the same PRN codes, the same navigation message format but carries whatever position and timing information the attacker wants the victim's receiver to compute. Because the attacker can transmit at a much higher power than the real satellite signals (which arrive attenuated after travelling 20,000 km), the receiver eventually locks onto the spoofed signals and discards the genuine ones.

The literature classifies spoofing attacks into three tiers of sophistication. The simplest approach, sometimes called meaconing, just captures real satellite signals and rebroadcasts them with a delay, causing the receiver to believe it is somewhere else in time. More capable attackers generate synthetic signals from scratch and transmit all visible satellites simultaneously, which is more convincing but still relatively easy to detect because the transition from the real position to the spoofed one is abrupt. The most dangerous variant is coherent spoofing, where the attacker first synchronizes the fake signals with the real ones so the receiver notices nothing unusual and then slowly drifts the spoofed position away from reality at a rate consistent with plausible vehicle motion. Unless the receiver is specifically designed to catch this kind of gradual deviation, it will follow the spoofed trajectory without any alarm.

3.2 Documented Real-World Cases

For a long time, GPS spoofing was treated as a theoretical threat by most of the navigation community. The events of 2017 in the Black Sea changed that perception. In June of that year, the US Maritime Administration received reports from multiple vessels near the Russian port of Novorossiysk that their GPS receivers were showing positions several kilometers inland, close to Gandhik Airport. The vessels were, of course, still at sea-the signals they were receiving were fake. The spoofing affected more than twenty ships over several days, and subsequent analysis suggested a

ground-based transmitter as the source, though attribution was never officially confirmed. Since then, the non-profit tracking site GPSJAM has catalogued persistent spoofing activity around several conflict zones, indicating that some actors now deploy this capability routinely rather than as a one-off stunt.

On the research side, Todd Humphreys' group at the University of Texas at Austin demonstrated in 2012 that a civilian drone flying a preprogrammed route could be redirected to an entirely different destination using a coherent spoofing attack, with the vehicle's on-board GPS reporting the spoofed trajectory as perfectly normal throughout the flight. The same team later showed that timing receivers used in power grids and telecommunications networks-which rely on GPS for highly accurate clock synchronization-could be fed false time signals, potentially destabilizing the frequency control of the grid. These demonstrations moved GPS spoofing firmly from the realm of speculation into that of demonstrated capability.

3.3 Countermeasures

There is no single silver bullet against spoofing, but several measures significantly raise the difficulty and cost of a successful attack. The most fundamental solution is signal authentication: embedding a cryptographic signature in the GNSS navigation message that allows the receiver to verify the signal's origin. The European Galileo system began rolling out its Open Service Navigation Message Authentication (OSNMA) service in 2023, and the US GPS programmed is evaluating a civilian authentication service called Chimera. Authentication does not prevent jamming, but it does defeat spoofing.

Cross-validation between GNSS and INS is another practical line of defense. A coherent spoofing attack that gradually shifts the reported position will produce an acceleration profile in the spoofed trajectory that may not match what the INS is measuring from the vehicle's actual motion. Comparing the two solutions and flagging divergences beyond a physically plausible threshold catches many attacks, even sophisticated ones. Using signals from multiple constellations-GPS plus Galileo plus BeiDou-raises the bar further, because an attacker must spoof all of them coherently. Finally, antenna-array techniques that examine the angle of arrival of signals can expose a terrestrial spoofer, since satellite signals come from above while a ground-based transmitter cannot perfectly fake that geometry.

4. Sensor Attacks

4.1 Why Sensors Are Vulnerable

It is tempting to think of sensor attacks as a niche concern compared to the more headline-grabbing problem of GPS spoofing, but I think that view underestimates how fundamental sensors are to the overall navigation solution-especially in the GPS-degraded environments that characterize most urban driving. If you can corrupt the INS, the LiDAR, or the camera inputs, you can effectively blind the vehicle to its true situation without ever touching the GNSS receiver.

Sensors are physically accessible in a way that software systems generally are not. They must interact with the real world-measuring acceleration, rotation, light, sound, radio waves-and that interaction creates channels through which a determined attacker can inject false information. The three main attack vectors are spoofing (feeding false stimuli to the sensor), jamming (overwhelming the sensor with noise so it cannot measure anything useful), and data injection (modifying the data stream between the sensor and its processing unit, for example by accessing the CAN bus).

4.2 Specific Attack Examples

One of the most striking sensor attack demonstrations I came across in preparing this paper was the acoustic attack on MEMS gyroscopes published by researchers at the University of Michigan and the University of South Carolina in 2017. MEMS (Micro-Electro-Mechanical Systems) gyroscopes contain tiny vibrating proof masses whose resonance frequency typically falls in the range of 20–30 kHz. The researchers showed that a loudspeaker placed near the gyroscope and driven at that resonance frequency could inject a controlled bias into the angular velocity output, effectively telling the vehicle's navigation system that it was rotating when it was not, or vice versa. They demonstrated that this was sufficient to make a self-balancing robot fall over and to cause a consumer drone to lose altitude control-without any direct physical contact with the device and without triggering any error flag in the firmware.

LiDAR spoofing presents a different kind of problem. A LiDAR scanner measures the time of flight of laser pulses to build a three-dimensional map of the environment; an attacker with a pulsed infrared laser synchronized to the scanner's own pulse timing can insert phantom returns that the scanner interprets as solid obstacles or can suppress real returns by blinding the detector. Research published at USENIX Security 2019 showed that off-the-shelf

components costing less than $60 were sufficient to create convincing phantom obstacles in the point cloud of a commercial autonomous vehicle LiDAR unit. A vehicle following its perception system's output would break for an obstacle that does not exist, or-more dangerously fail to detect one that does.

Camera-based perception systems face the challenge of adversarial examples: carefully designed visual patterns that cause a deep neural network to misclassify objects with high confidence, while appearing innocuous to a human observer. Ekholm et al. (2018) demonstrated that adding specific sticker patterns to a stop sign caused a state-of-the-art object classifier to identify it as a 45 mph speed limit sign in 100% of their physical test cases. This is not a theoretical concern-it is a practical attack that requires only a printer and some tape.

4.3 Countermeasures

The most robust defense against sensor attacks is the same principle that makes aircraft safe: redundancy and cross-checking. If the navigation solution is derived from multiple independent sensor modalities GNSS, INS, camera, LiDAR, radar then corrupting one of them is not sufficient to corrupt the output, provided the fusion algorithm is designed to detect and down-weight inconsistent inputs. Anomaly detection algorithms that monitor each sensor's output for physically implausible values or sudden discontinuities can catch many attack patterns, particularly the kind of abrupt spike that simple jamming or unsophisticated spoofing produces. More sophisticated attacks require machine learning models trained on large datasets of normal sensor behavior, which can detect subtle statistical anomalies that threshold-based checks miss. On the hardware side, shielding sensors from electromagnetic and acoustic interference, and encrypting the data path between the sensor and its processing unit, reduces the available attack surface considerably.

5. Encryption in Navigation Systems

5.1 The Problem with Unencrypted Navigation Data

A modern connected vehicle is continuously exchanging navigation-relevant data with the outside world: differential GNSS corrections from a cloud service to improve positioning accuracy, high-definition map tiles that may include precise lane geometry and speed limit data, software and firmware updates that affect the behavior of the navigation and control systems, and real-time V2X messages about hazards, traffic, and roadworks. Without encryption, any of these exchanges can be intercepted, read, and-

more dangerously-modified by an attacker positioned anywhere on the network path between the vehicle and the server.

The consequences of unencrypted navigation data go beyond privacy. An attacker who can modify the map tiles a vehicle receives could, in principle, add phantom road closures that divert the vehicle onto a different route, remove speed limit markings that the automated speed control system relies on, or alter the geometry of an intersection in ways that cause a self-driving system to misplace itself. An attacker who can replay an old firmware update could roll back security patches that have been applied to fix known vulnerabilities. These are not far-fetched scenarios; they are logical extensions of attack types that are already well understood in enterprise IT security, applied to a new context.

5.2 Encryption Techniques Used in Practice

The standard approach is to use symmetric encryption for bulk data transfer and asymmetric encryption for key exchange. AES (Advanced Encryption Standard) with 128- or 256-bit keys is the workhorse for symmetric encryption; it is fast enough to run on embedded processors without imposing significant latency, and it is computationally secure against any known classical attack. For key exchange, the trend has shifted from RSA (which requires relatively large key sizes-at least 2048 bits-to be secure) towards Elliptic Curve Cryptography (ECC), which achieves equivalent security with much shorter keys: a 256-bit ECC key is considered roughly equivalent to RSA-3072. This matters in the context of embedded navigation systems because shorter keys mean faster computation and smaller memory footprint.

A Public Key Infrastructure (PKI) ties these cryptographic tools together by managing the certificates that bind a public key to a specific device identity. In the vehicle context, a PKI might be operated by the vehicle manufacturer or by a shared industry body, and every ECU or telematics unit in the vehicle would have a certificate issued by that authority. This allows different components of the system-and external entities like V2X roadside units or cloud servers-to verify each other's identity before exchanging sensitive data. The challenge, as I will discuss shortly, is that managing certificates across millions of vehicles over their full service life is not a trivial operational problem.

For the GNSS signal itself, military users of GPS have long had access to the encrypted P(Y) code, which is resistant to spoofing because an attacker cannot generate a valid P(Y) signal without knowing the classified encryption key. Civilian users are only now

gaining access to analogous protections. Galileo's OSNMA service, which reached initial operational capability in 2023, appends a cryptographic message authentication code to the navigation message; a receiver with the correct public key can verify that the message was signed by a genuine Galileo ground control station.

5.3 Key Challenges

Three problems come up repeatedly in the literature on encryption for navigation systems, and they are worth discussing in some detail because they explain why, despite the fact that strong cryptographic tools have existed for decades, many deployed systems still lack adequate encryption.

The first is the latency-security trade-off. Encryption is not free: a TLS handshake takes time, and computing a digital signature or verifying a certificate requires processor cycles that an embedded ECU might not have available if it is already running a real-time control loop. For a navigation server in a data centre, this overhead is negligible. For a processor in an autonomous vehicle that needs to update its position estimate and issue control commands every 10 milliseconds, even a few milliseconds of cryptographic overhead can be significant. Lightweight cryptography standards-the suite that NIST finalized in 2023 under its Lightweight Cryptography (LWC) program-are specifically designed to address this, but they have not yet been widely adopted.

The second problem is key management at scale. A vehicle manufacturer deploying hundreds of thousands of vehicles per year, each with multiple ECUs that each need their own certificates, is looking at a key management operation of considerable complexity. Keys need to be securely generated and injected during manufacturing, stored safely on the vehicle, renewed before they expire, and revoked if a component is compromised. Getting any one of these steps wrong can undermine the entire security of the system.

The third challenge is the legacy problem. Vehicles on the road today have service lives of fifteen to twenty years, and many of them were designed before strong encryption was considered a requirement for navigation and telematics. Retrofitting modern cryptographic support onto systems that were not designed for it ranges from difficult to impossible, which means that the overall fleet security improves only gradually as older vehicles age out.

6. Secure Communication Protocols

6.1 TLS and IPsec

For vehicle-to-cloud communication-map updates, traffic data, software over-the-air updates-TLS 1.3 has become the standard approach, for the same reason it dominates web security: it is well-analyzed, widely implemented, and provides a strong combination of confidentiality, integrity, and authentication. The TLS 1.3 handshake is notably faster than its predecessors, requiring only one round trip (versus two for TLS 1.2), which is important for mobile clients that may have intermittent connectivity. Session resumption allows a vehicle that briefly loses connectivity to re-establish a secure channel without a full handshake, reducing the latency cost further.

IPsec operates at the IP layer rather than the transport layer, which makes it suitable for securing communication between infrastructure components-for example, between a fleet management center and roadside V2X units-where the application protocols themselves may not support encryption. In Tunnel Mode, IPsec encapsulates the entire IP packet, including the original headers, which prevents an eavesdropper from learning even which infrastructure nodes are communicating with each other.

6.2 V2X Security Protocols

Vehicle-to-Vehicle and Vehicle-to-Infrastructure communication presents somewhat different requirements from vehicle-to-cloud. Messages must be verified in near-real time-a vehicle receiving a hazard alert from another vehicle needs to trust that message within a few milliseconds, not after completing a full certificate validation chain. The approach taken in both the DSRC (IEEE 802.11p) and C-V2X (3GPP Release 14 and later) standards is to use short-lived pseudonymous certificates: each vehicle holds a pool of certificates, each valid for only a few minutes, and switches between them regularly to prevent long-term location tracking by correlation of signed messages. The certificates are pre-signed by a trusted authority, so verification at reception is fast-it requires only a signature check rather than a full handshake.

UAV fleets and vehicular ad hoc networks (VANETs) add the complication of rapidly changing network topology: nodes join and leave the network continuously, routing paths change from second to second, and there may be no reliable infrastructure backbone to fall back on. Secure extensions to AODV and OLSR routing protocols have been proposed that use digital signatures to authenticate route announcements and detect malicious nodes that are advertising false routes; the challenge is doing this with low

enough overhead that the routing protocol can still converge quickly enough to be useful.

6.3 Implementation Challenges

The gap between the security protocols that exist on paper and what is actually deployed in production vehicles is, in my view, one of the most underappreciated problems in this field. Several factors contribute to it. Automotive development cycles are long-typically four to six years from concept to production-and security standards evolve faster than that; a vehicle that was designed to the state of the art in 2020 may be shipped with protocols that are already considered outdated. Many ECU suppliers treat cryptographic support as an optional extra rather than a baseline requirement, which means that manufacturers face a fragmented landscape of components with very different security capabilities. And the integration testing required to ensure that security mechanisms work correctly across a complex network of dozens of ECUs from different suppliers is expensive and time-consuming, creating pressure to cut corners.

7. Integrated Security Approaches

The previous four sections have treated GPS spoofing, sensor attacks, encryption, and communication security as separate topics, which is a useful analytical framework but a somewhat misleading picture of how attacks actually work. In practice, a sophisticated attacker does not pick one vulnerability and ignore all the others; they look for a combination of weaknesses that can be chained together. A spoofing attack is more effective if the vehicle's INS cross-check has been disabled by a prior sensor attack; a data injection attack is easier if the communication channel is not encrypted. This is why a point-by-point approach to security-fixing each vulnerability separately as it is discovered-tends to lag behind the threat rather than get ahead of it.

The concept of defense in Depth, borrowed from military strategy and well-established in enterprise IT security, provides a more coherent framework. The idea is to implement multiple independent layers of protection, each assuming that the layers above and below it may fail. Applied to vehicle navigation, this means security controls at the physical layer (shielded sensors, directional anti-jamming antennas), the signal layer (GNSS authentication, carrier-to-noise monitoring), the data layer (encryption, cryptographic integrity checks), the network layer (secure protocols, firewalls, intrusion detection), and the application

layer (cross-sensor validation, AI-based anomaly detection, safe-mode fallback behaviors). A successful attack must penetrate all of these layers, which is substantially harder than breaching any one of them.

The GNSS-INS combination deserves particular mention as an example of integrated security. Neither system is secure on its own: GNSS can be spoofed, and INS drifts over time. But together they provide mutual cross-validation that makes attacks significantly harder. A coherent GPS spoofing attack that slowly shifts the reported position will produce a velocity profile inconsistent with what the INS is integrating from the vehicle's actual motion; a Kalman filter designed to catch exactly this kind of divergence will flag the inconsistency long before the navigation error becomes safety-critical.

Continuous monitoring and rapid response are also essential elements of an integrated approach. Static security mechanisms-encryption algorithms, firmware access controls-are necessary but not sufficient; new vulnerabilities will inevitably be discovered, and the ability to detect anomalies in real time and deploy patches quickly is what determines whether a vulnerability can be exploited at scale before it is fixed. Secure over-the-air update mechanisms, which themselves must be protected against the very attacks they are designed to remediate, are therefore a critical piece of the overall architecture.

8. Future Trends

8.1 Post-Quantum Cryptography

The possibility that large-scale quantum computers will eventually be able to break RSA and ECC has been discussed in the cryptography community for over twenty years, but it has acquired new urgency recently for two reasons. First, NIST finalised its first post-quantum cryptography (PQC) standards in 2024-CRYSTALS-Kyber for key encapsulation and CRYSTALS-Di lithium for digital signatures-giving industry concrete algorithms to migrate to. Second, the 'harvest now, decrypt later' threat has become more credible: an adversary with sufficient resources can record encrypted traffic today and decrypt it once quantum hardware matures, which means that data with a secrecy requirement extending more than a decade into the future-military navigation data being an obvious example-should arguably be protected with quantum-resistant algorithms now.

For vehicle navigation specifically, the migration to PQC is

complicated by the same legacy and resource constraints discussed earlier in the context of encryption. Post-quantum algorithms tend to have larger key sizes and signature sizes than their classical equivalents, which increases bandwidth and memory requirements; they also tend to be computationally heavier. Whether lightweight variants suitable for constrained embedded processors can be standardized and deployed before quantum computers become a practical threat is an open question.

8.2 AI-Based Threat Detection

Machine learning is increasingly attractive for navigation security because it can detect attack patterns that are too subtle or too novel for rule-based systems to catch. For GPS spoofing specifically, neural network classifiers trained on receiver independent exchange (RINEX) observation files-the standard format for logging raw GNSS measurements-have demonstrated detection accuracy above 95% in controlled tests, even for coherent spoofing attacks that produce minimal visible artefacts. Deep learning models trained on sensor telemetry can similarly identify the statistical signatures of acoustic gyroscope attacks or LiDAR spoofing that threshold-based checks would miss entirely.

The concern I have with AI-based detection is adversarial robustness. An attacker who knows that a detection model is deployed can in principle probe the model's decision boundary and craft attacks that evade it-the same adversarial example problem that affects camera-based perception, turned against the security system itself. Defending against adversarial attacks on the detector requires techniques like ensemble methods, adversarial training, and certified defences that are active research areas but not yet mature enough for straightforward deployment. This does not mean AI-based detection should not be used-it clearly adds value-but it should be treated as one layer of a defence-in-depth strategy rather than a replacement for cryptographic guarantees.

8.3 Blockchain for Data Integrity

Blockchain has attracted considerable interest as a mechanism for ensuring the integrity of shared navigation data-HD maps, traffic events, road condition reports-in environments where multiple independent parties contribute data and no single entity is trusted by all the others. A permissioned blockchain among road operators, vehicle manufacturers, and regulators could in principle provide cryptographic proof that a map update was submitted by an authorised source, has not been modified in transit, and has been seen and validated by a quorum of participants before being

accepted by vehicles.

The practical obstacles are significant. Current blockchain systems are too slow and too computationally expensive for real-time safety-critical messages; the consensus mechanisms that provide tamper-evidence also introduce latency that is incompatible with the millisecond response times that autonomous driving requires. The more promising application is for slower-moving data-periodic map updates, software release records, certificate revocation lists-where the latency overhead is acceptable and the tamper-evidence property is genuinely valuable.

8.4 5G and Mobile Edge Computing

Fifth-generation cellular networks bring two improvements that are directly relevant to vehicle navigation security. The first is latency: 5G's design target of sub-1 ms round-trip time in urban deployments makes it, for the first time, fast enough to carry safety-critical V2X messages that were previously only feasible over dedicated short-range DSRC links. The second is the 5G security architecture itself, which includes mutual authentication between the device and the network (preventing fake base station attacks that affected earlier cellular generations), user-plane integrity protection, and subscriber identity concealment.

Mobile Edge Computing (MEC), which moves computation from centralized data centers to servers co-located with 5G base stations, reduces both the latency and the attack surface of cloud-connected navigation. Processing that currently has to travel hundreds of milliseconds to a distant server can instead be offloaded to an MEC node a few hundred meters away; this also means that sensitive position data travels a much shorter path through the network, reducing the number of potential interception points.

9. Conclusion

The central argument of this paper is that cybersecurity in vehicle navigation is not a problem that admits a neat technical solution. The vulnerabilities are structural: civilian GNSS signals were designed for accuracy and availability, not for security; sensors must be physically exposed to function; communication links must be open enough to exchange real-time data. Each of these design choices was reasonable given the constraints and threat model of the time, but the threat model has changed.

GPS spoofing attacks have moved from academic

demonstrations to documented operational use. Sensor attacks that seemed exotic in a university lab are now well within the reach of motivated non-state actors with modest resources. The regulatory and liability pressure on autonomous vehicle manufacturers to demonstrate that their systems are secure will only increase as deployment scales up.

The good news is that the countermeasures are also maturing. Signal authentication services for civilian GNSS are being deployed. Post-quantum cryptography standards have been finalized. AI-based anomaly detection is improving. The 5G security architecture addresses several weaknesses of earlier cellular standards. What remains to be done is the hard engineering and organizational work of deploying these measures consistently, managing the legacy fleet, and treating security as a first-class design requirement rather than an afterthought. None of that is easy, but the alternative large-scale fleet of connected autonomous vehicles with inadequately secured navigation systems-is a risk that society cannot afford.

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